Replay changes sampling and reward scale

Control and experiment

Baseline L2_t1 (d807d91d): temperature1, no replay,128 fresh prompts. BCR16_T1 (73acea80):112 fresh + up to16 replay,400-prompt buffer,worst greedy word score,seed42. Replay regenerates outputs using current policy.

Reward and shared settings

Reward divisor4 →3; same component weights and clip[-2,2]. This scales the same weighted sum by4/3. Both: ReMax,Qwen9B,LoRA320,LR3e-6 constant,8 rollouts,16 GPUs,two epochs; validation temperature0/n1.

Provenance and limits

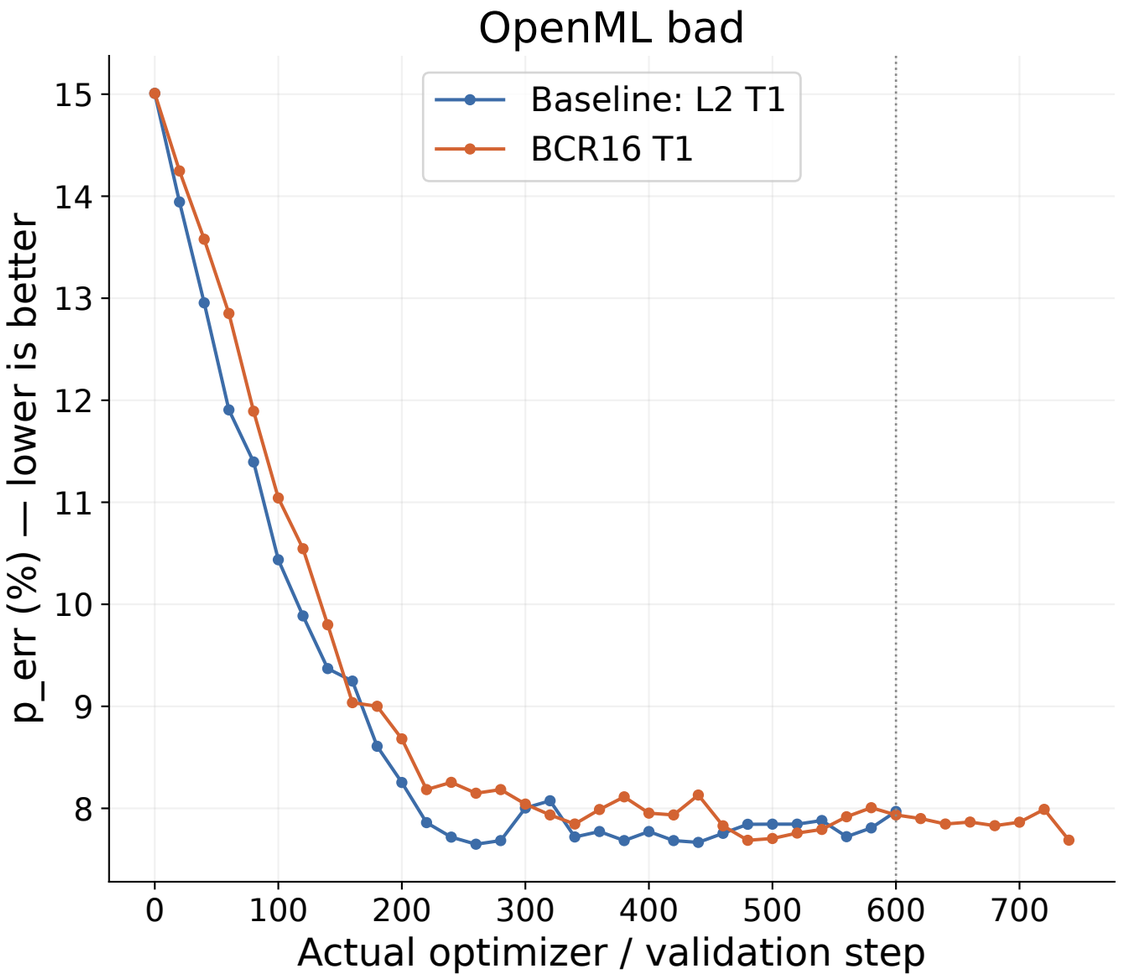
Baseline03680b38 →target1d72d5a. Scorer,normalizer,actor loss and core advantage code identical; replay integration changes trainer input sampling. Multiple recoveries reset replay; equal steps have different fresh-data exposure.

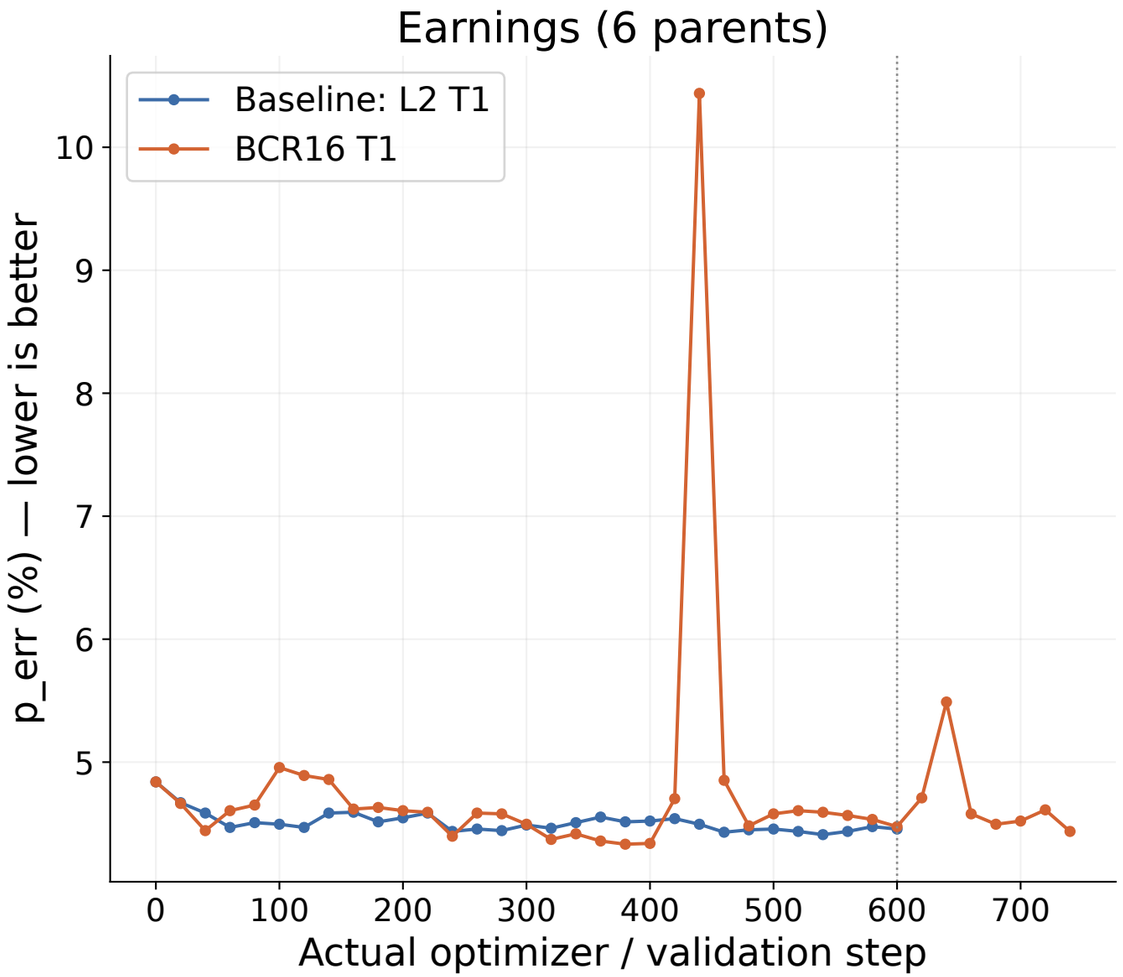
Training stopped

User stop confirmed by Ray: recovery750/driver10000000 STOPPED,0 allocated GPUs. Monitor deleted. Historical last confirmed optimizer755; recovery750 stopped during initial validation with no new optimizer update.

Pinned source audit and collected W&B configs; live stop verified by exact Ray submission.

Validation trajectories — preserve dataset tradeoffs





initial: 15.008% @0 / latest: 7.969% @600 / best: 7.649% @260

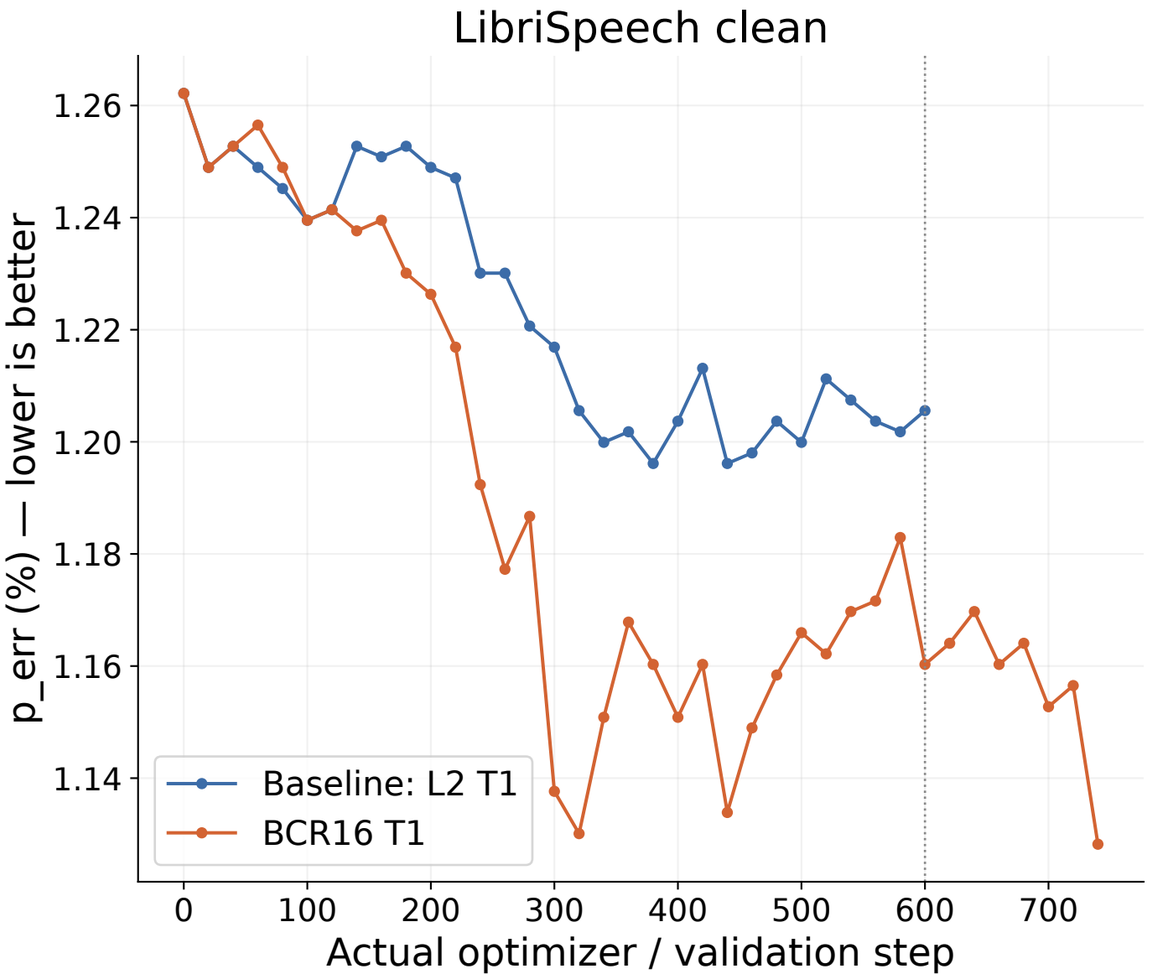
initial: 15.008% @0 / latest: 7.688% @740 / best: 7.687% @480

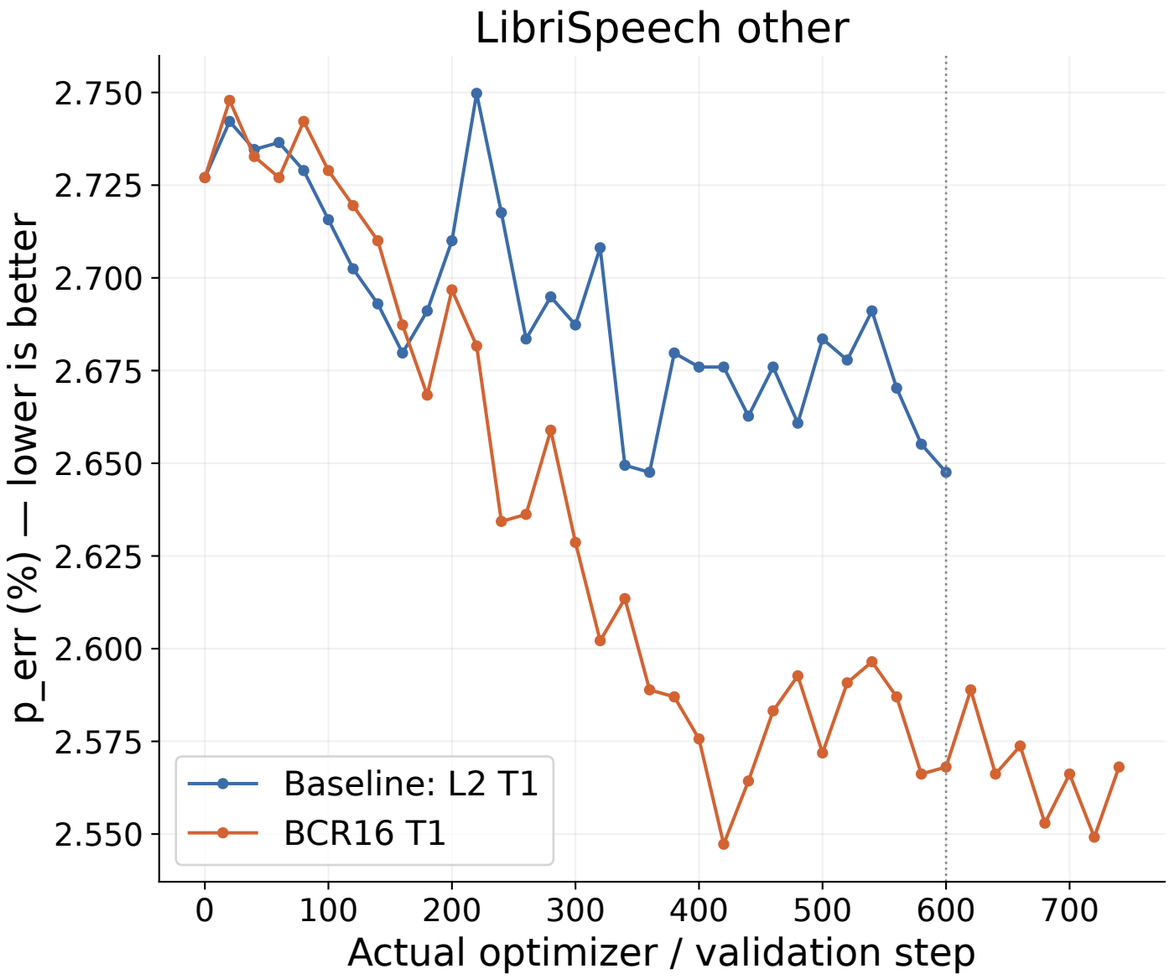
initial: 4.839% @0 / latest: 4.456% @600 / best: 4.411% @540

initial: 4.839% @0 / latest: 4.437% @740 / best: 4.333% @380

Observed points only; lines are guides. Target latest740 vs baseline600 is an unequal-budget comparison. Initial/latest/best shown separately.

Validation trajectories — preserve dataset tradeoffs





initial: 1.262% @0 / latest: 1.206% @600 / best: 1.196% @380

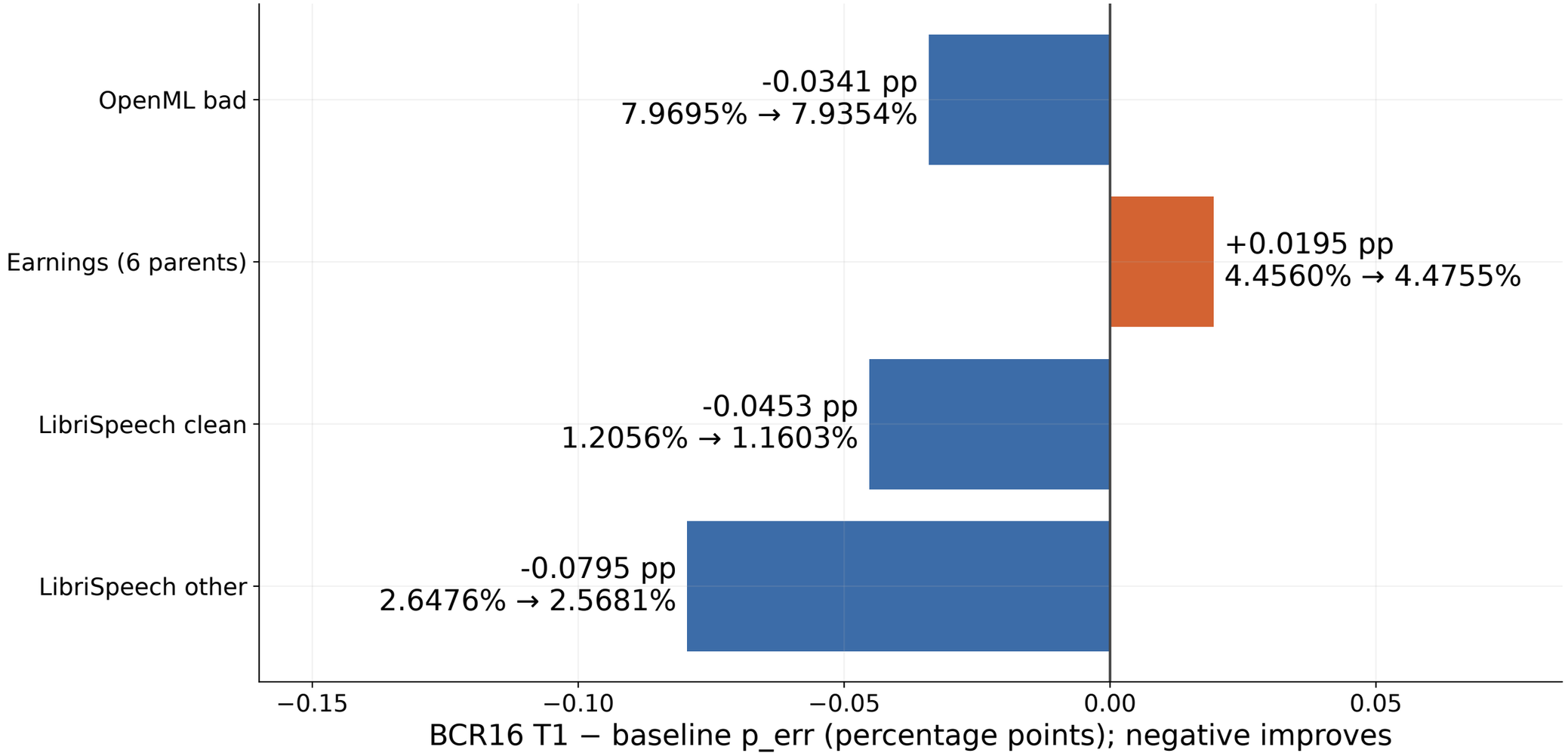
initial: 1.262% @0 / latest: 1.128% @740 / best: 1.128% @740

initial: 2.727% @0 / latest: 2.648% @600 / best: 2.648% @360

initial: 2.727% @0 / latest: 2.568% @740 / best: 2.547% @420

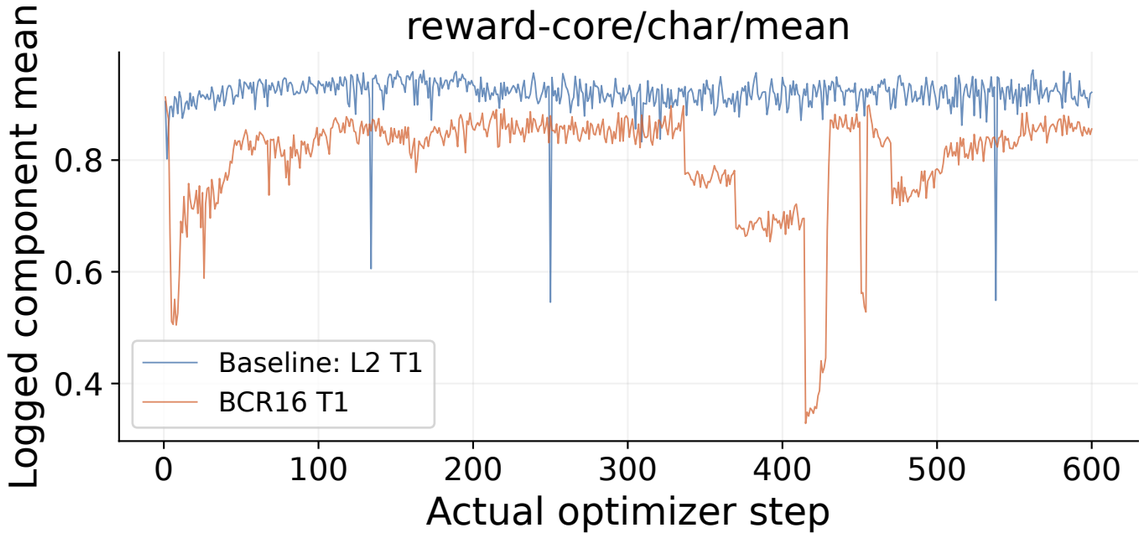
Observed points only; lines are guides. Target latest740 vs baseline600 is an unequal-budget comparison. Initial/latest/best shown separately.

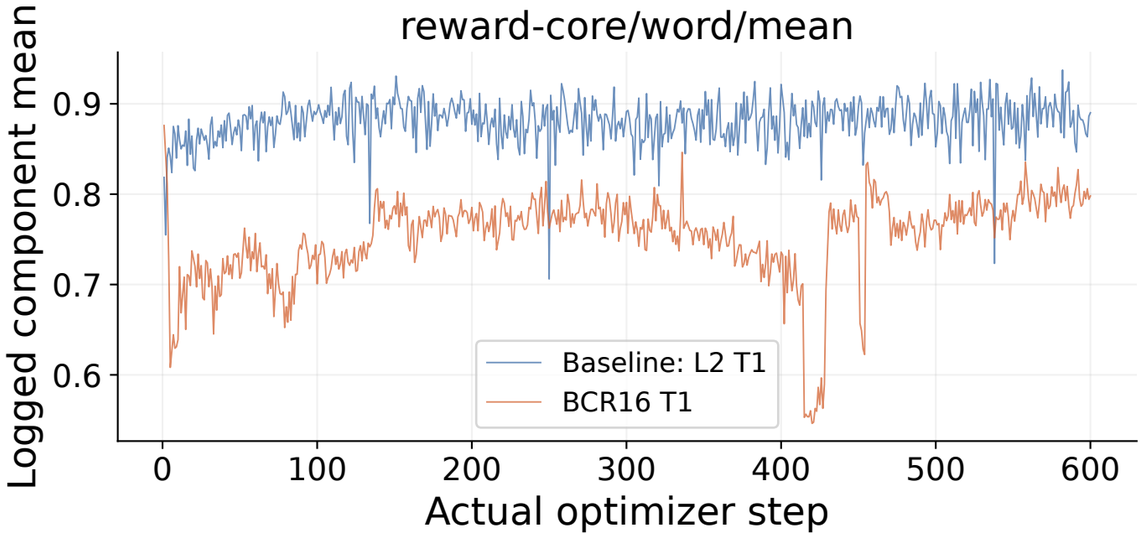
At step600: LS gains; Earnings slightly regresses

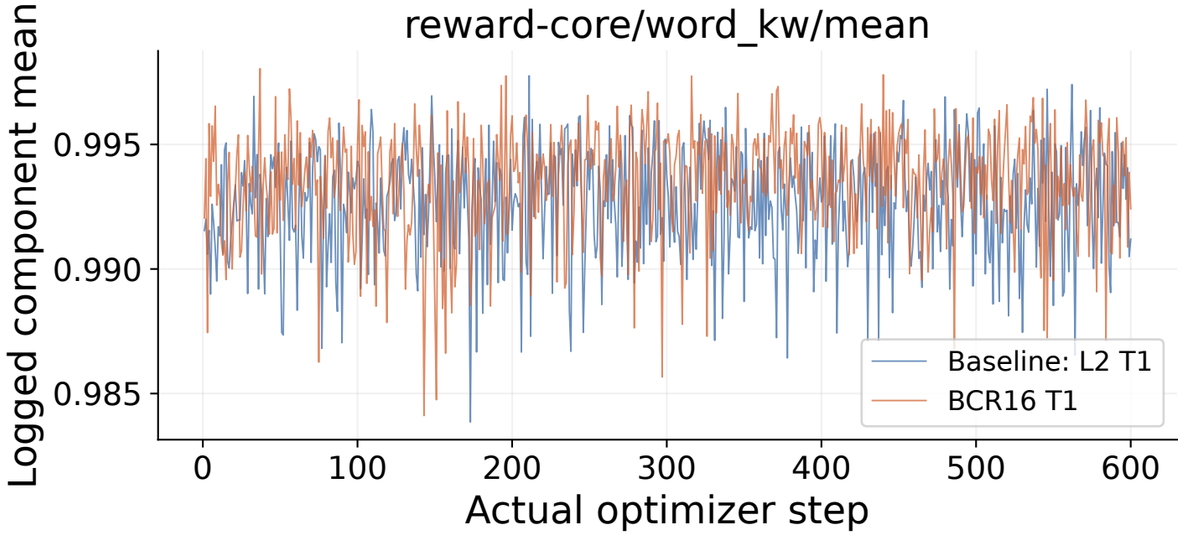


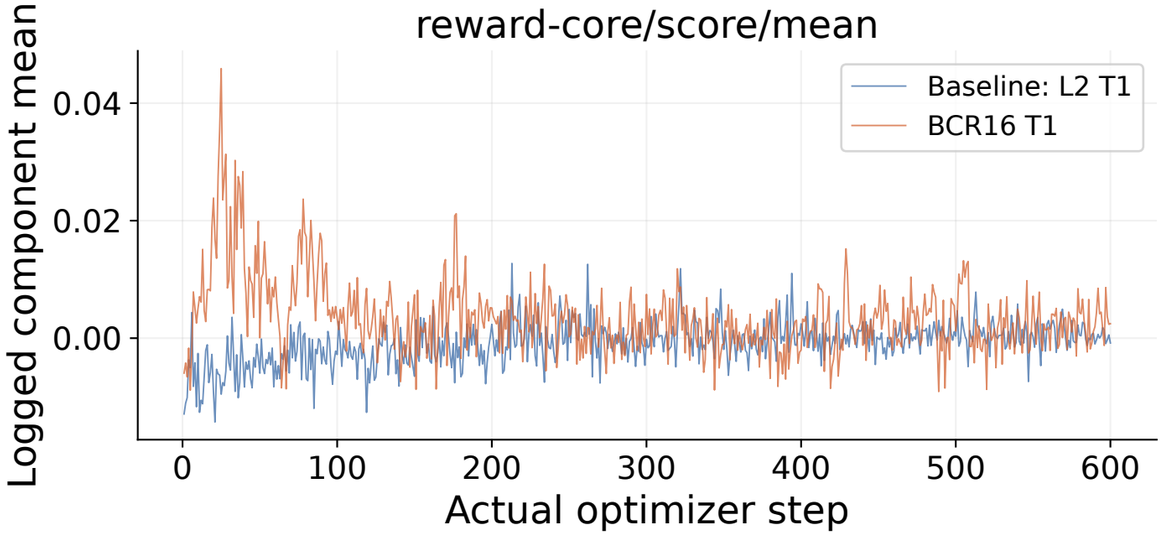
Exact shared step600. No cross-dataset averaging; single-run descriptive comparison with reward-scale and exposure confounds.

reward-core: raw batch means through step600



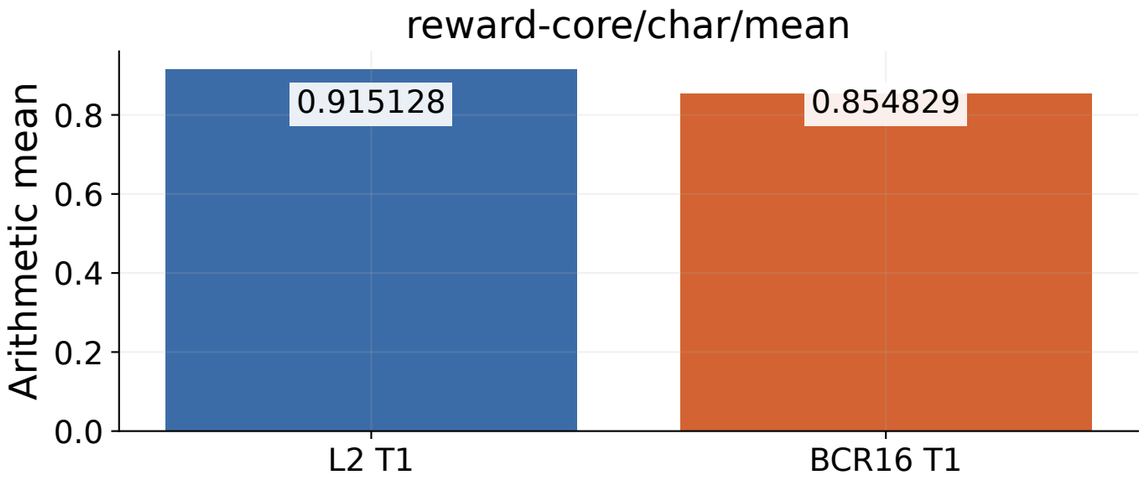


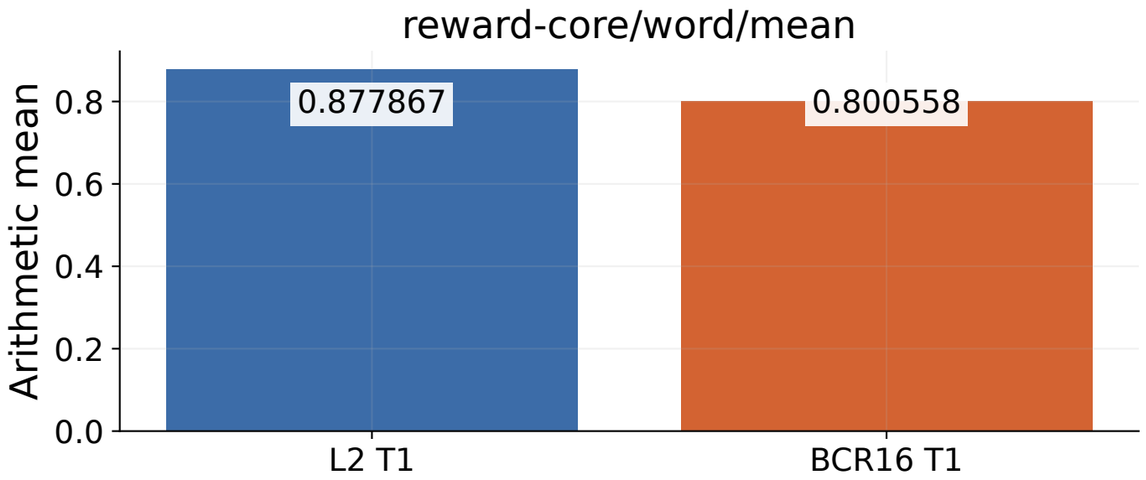


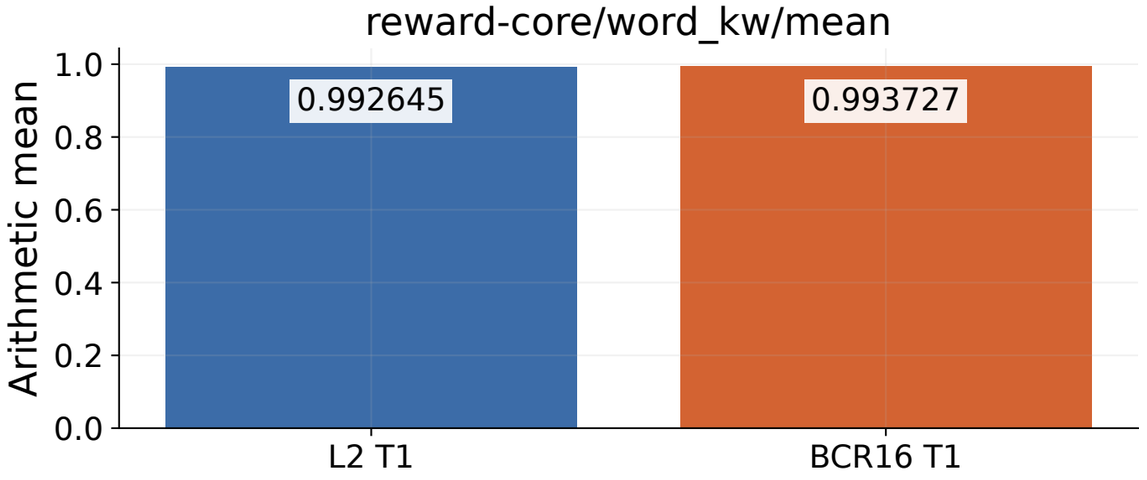


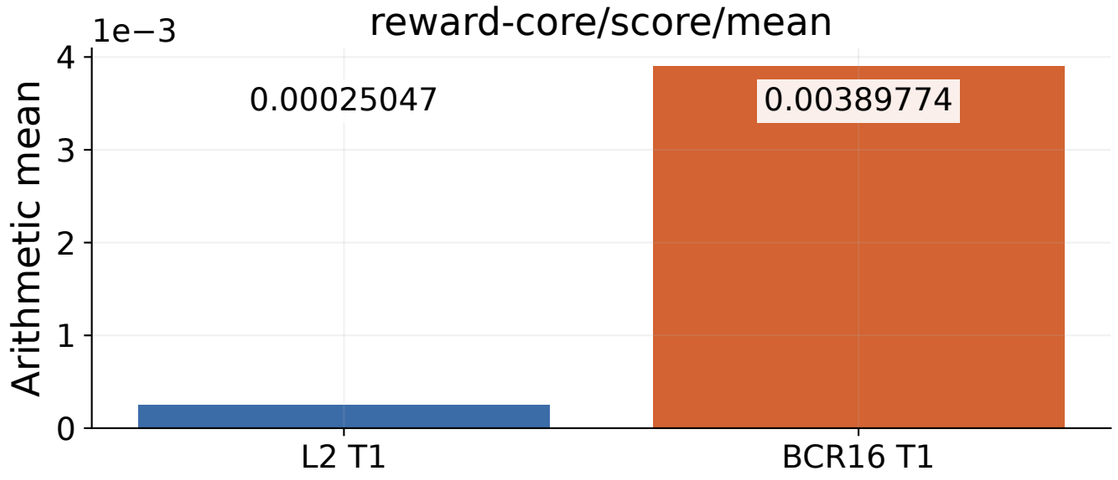
No smoothing/interpolation. Logged history includes abandoned pre-recovery updates; all22 keys retained. Batches differ by replay.

reward-core: exact shared window591–600



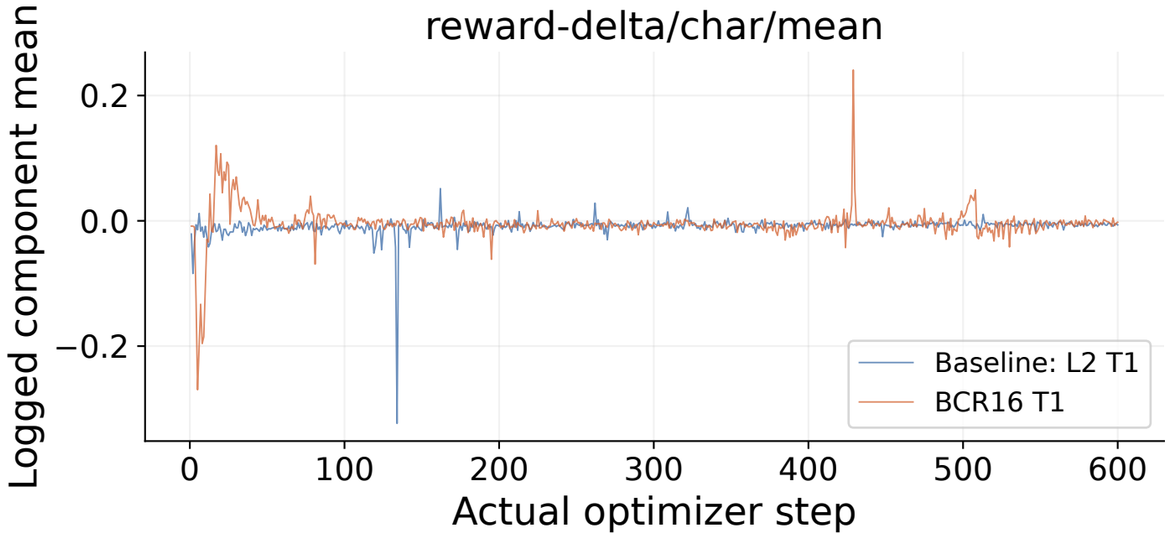


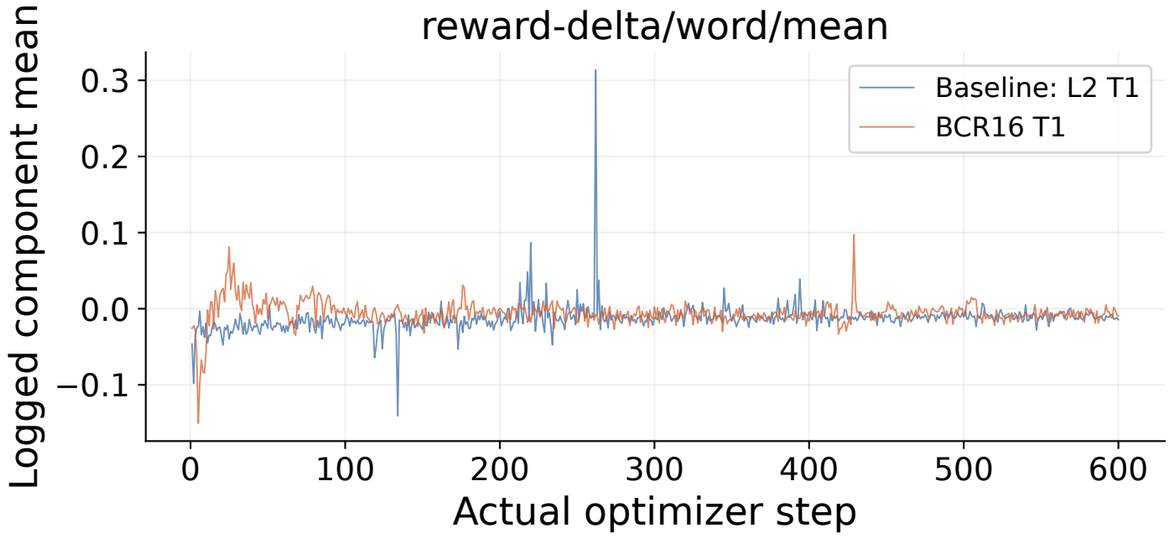


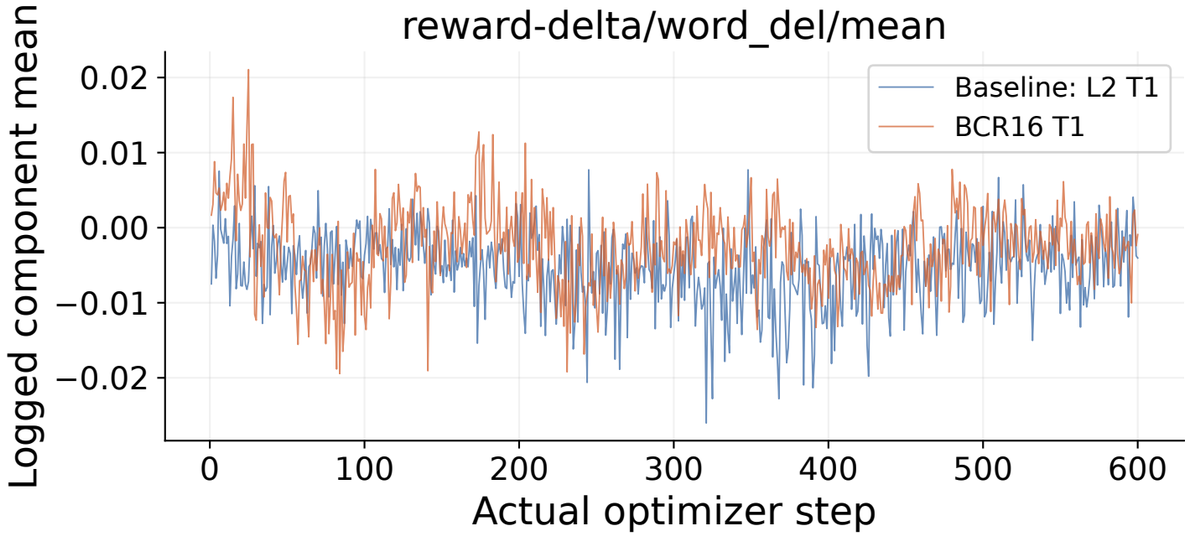


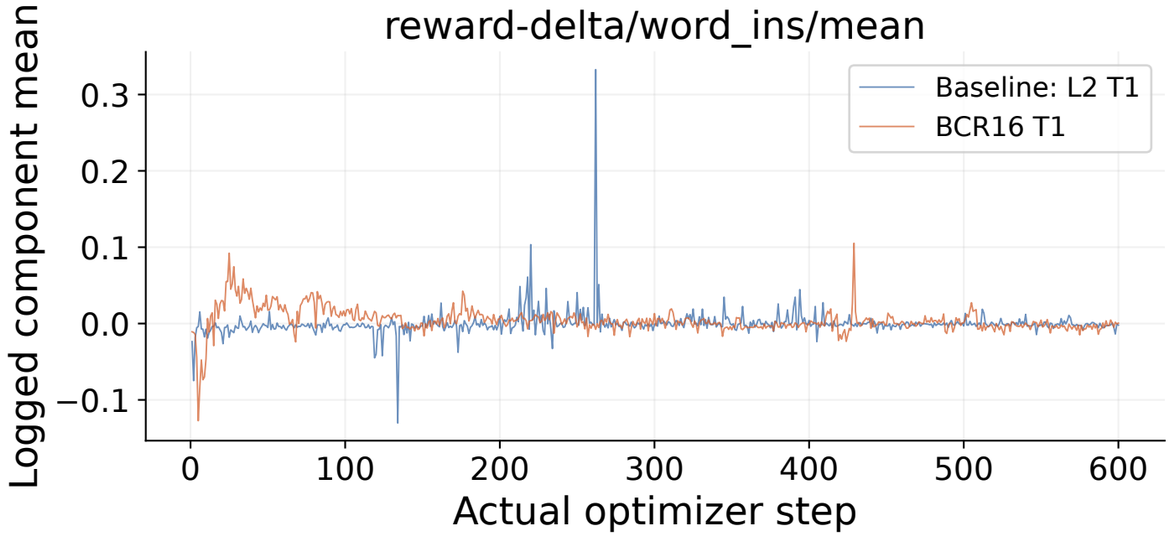
10 observations per run per component; arithmetic means. Core score has a different divisor and is not a like-for-like quality metric.

reward-delta: raw batch means through step600



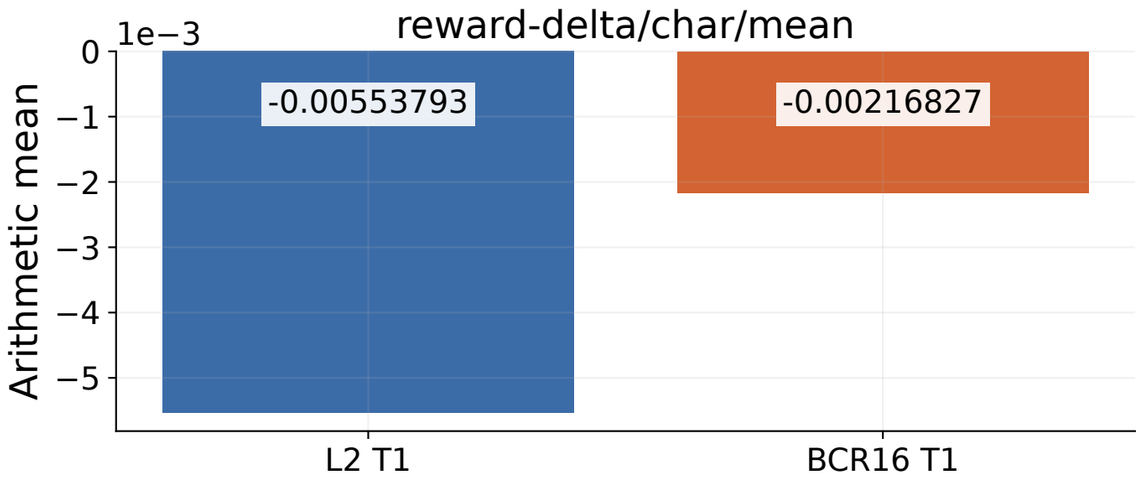


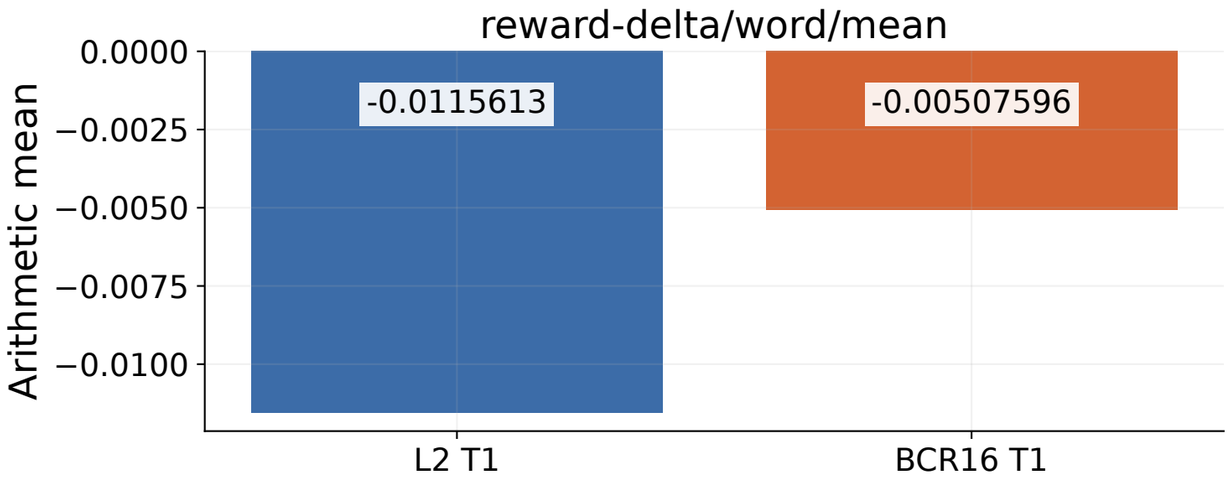


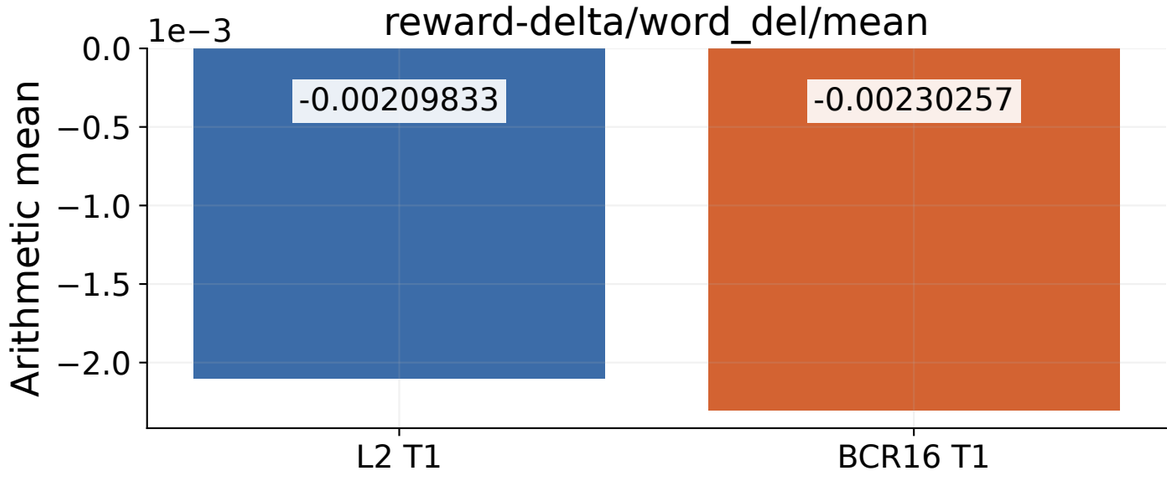


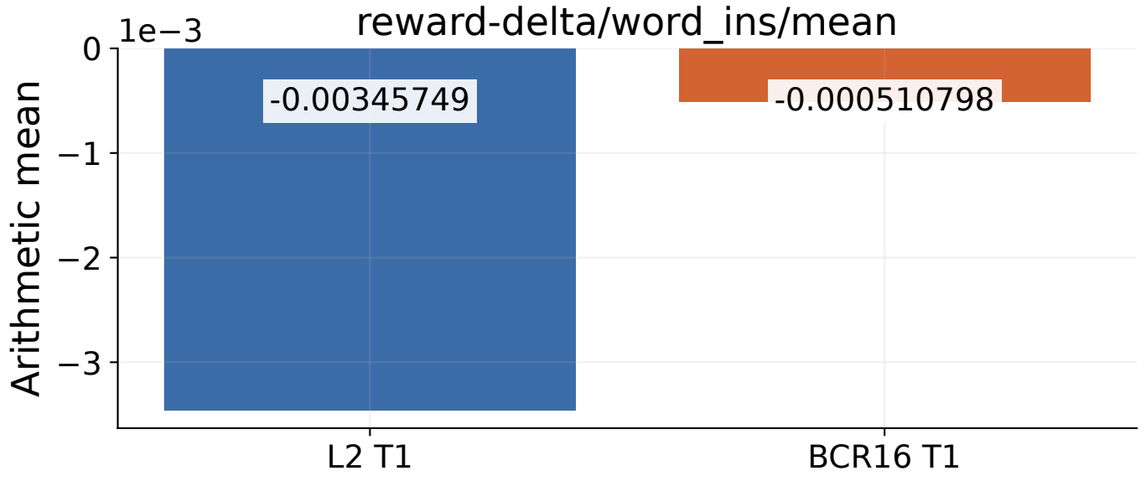
No smoothing/interpolation. Logged history includes abandoned pre-recovery updates; all22 keys retained. Batches differ by replay.

reward-delta: exact shared window591–600



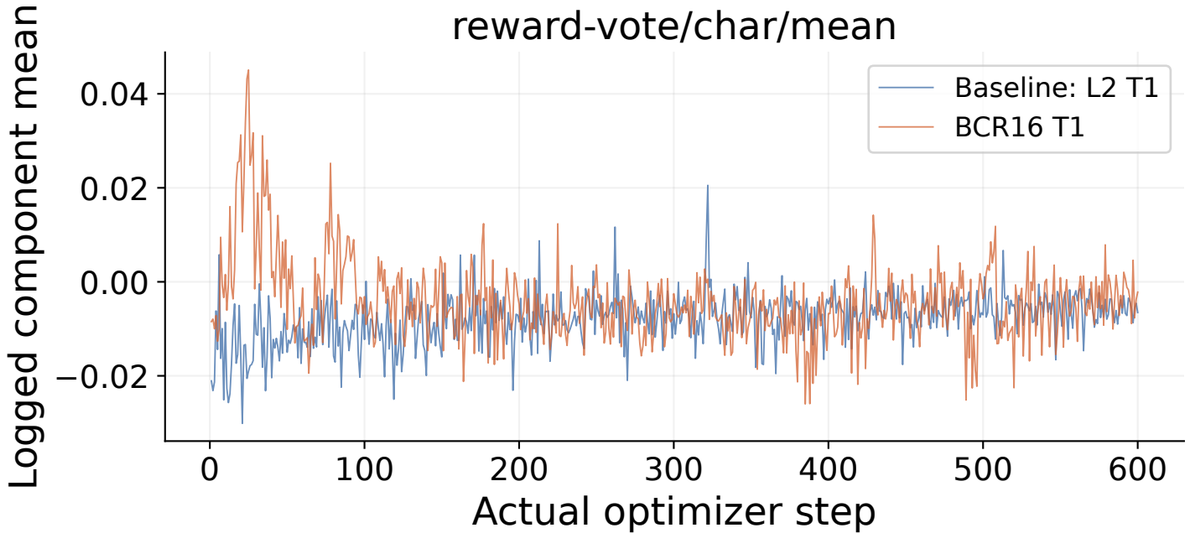


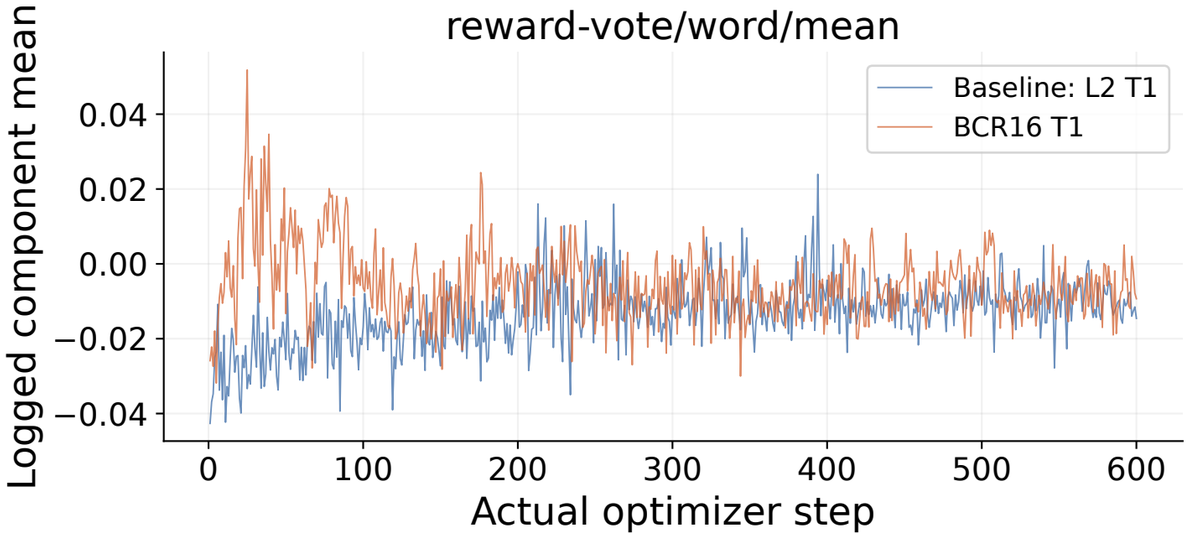


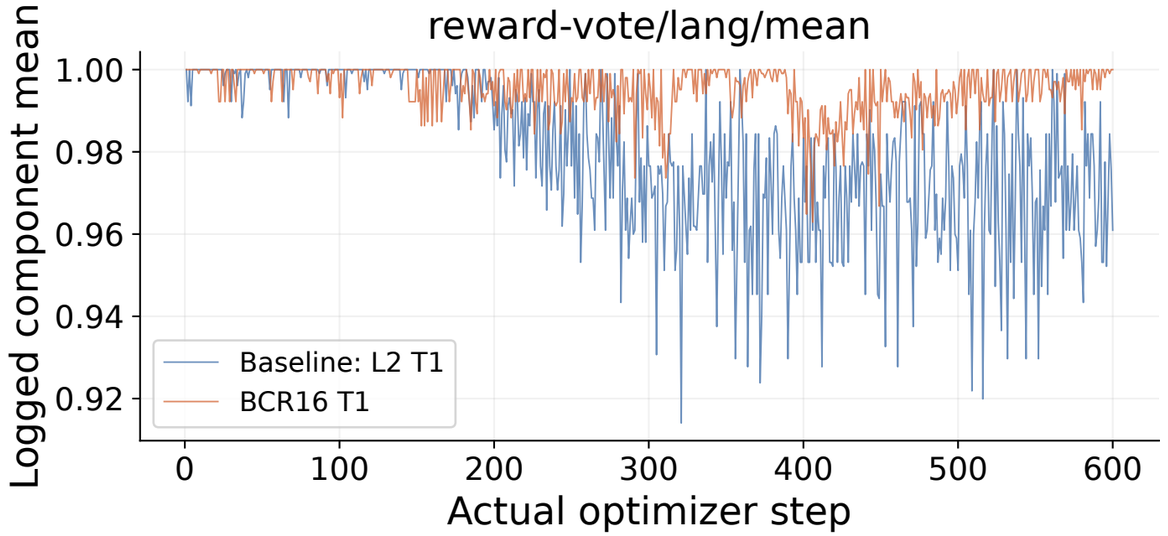


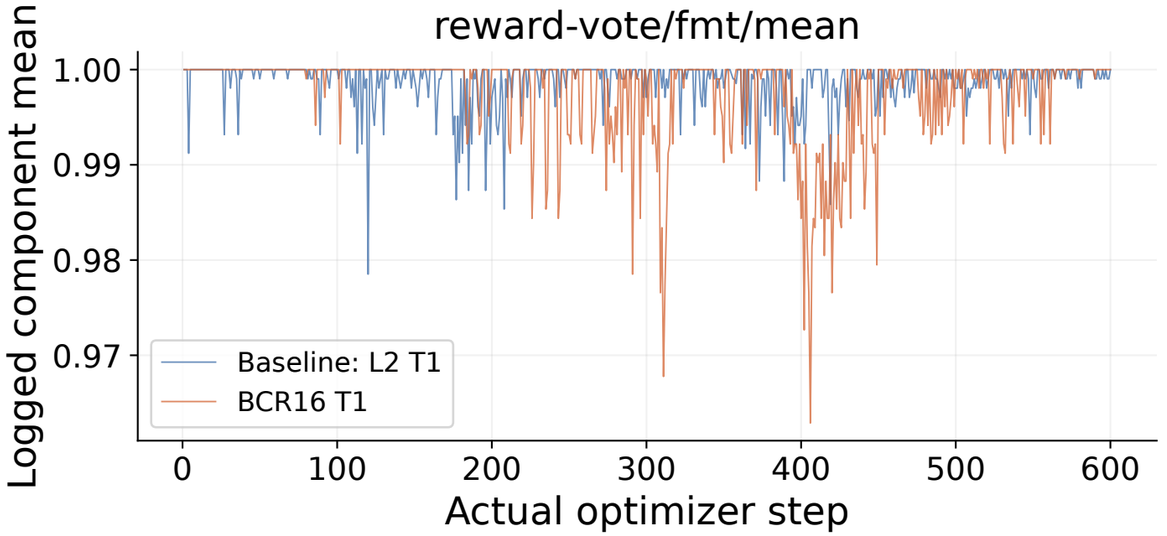
10 observations per run per component; arithmetic means. Core score has a different divisor and is not a like-for-like quality metric.

reward-vote: raw batch means through step600



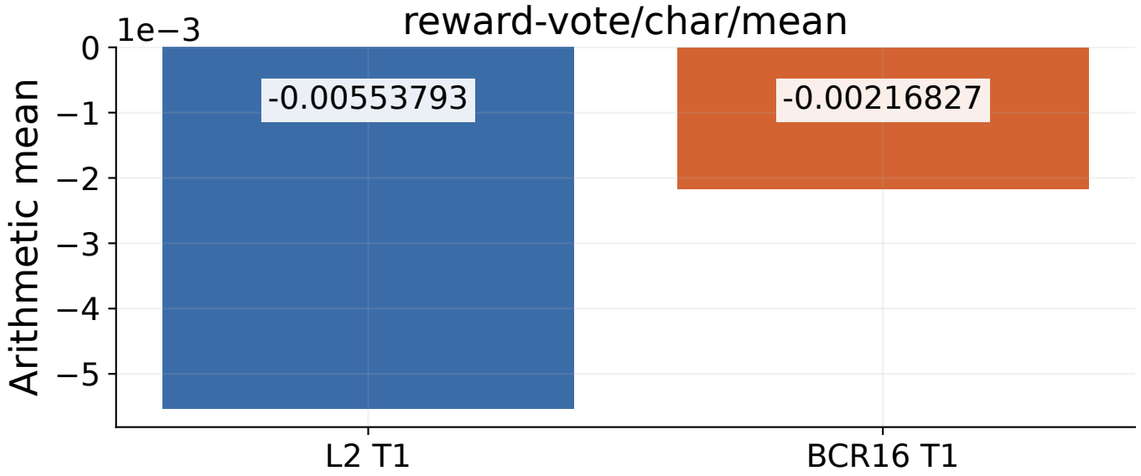


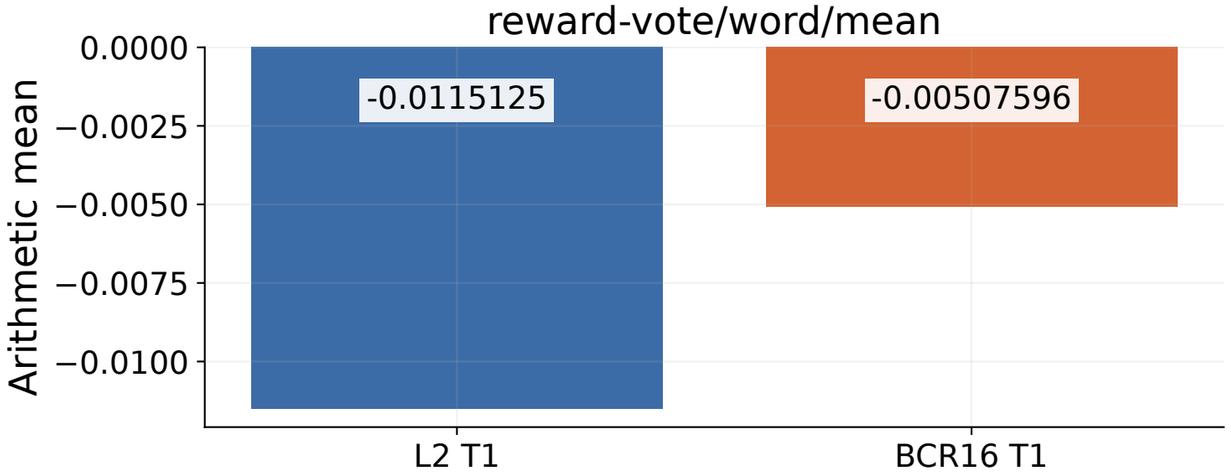


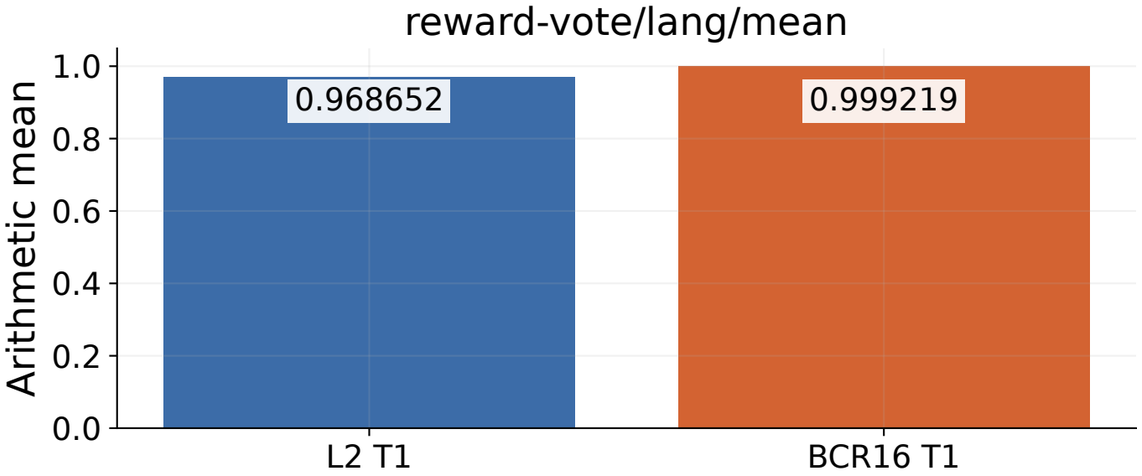


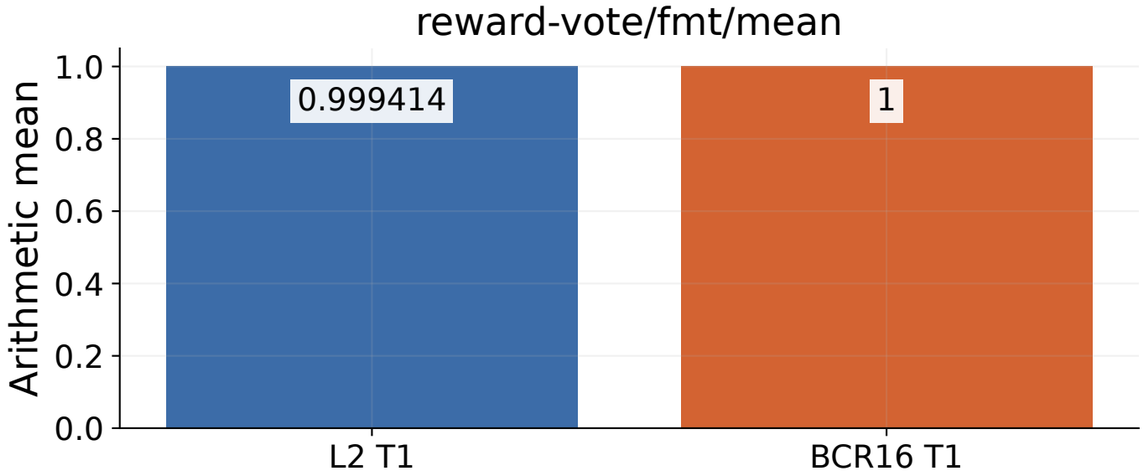
No smoothing/interpolation. Logged history includes abandoned pre-recovery updates; all22 keys retained. Batches differ by replay.

reward-vote: exact shared window591–600



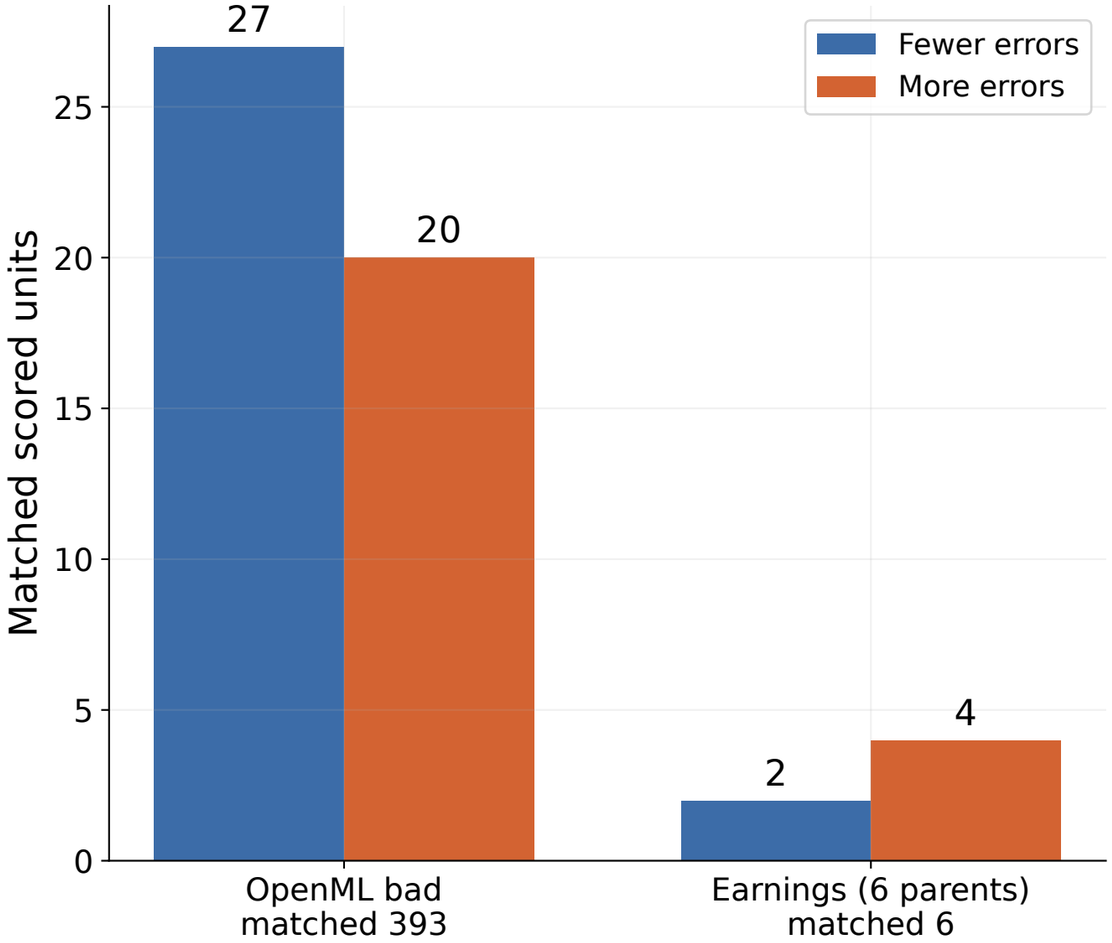


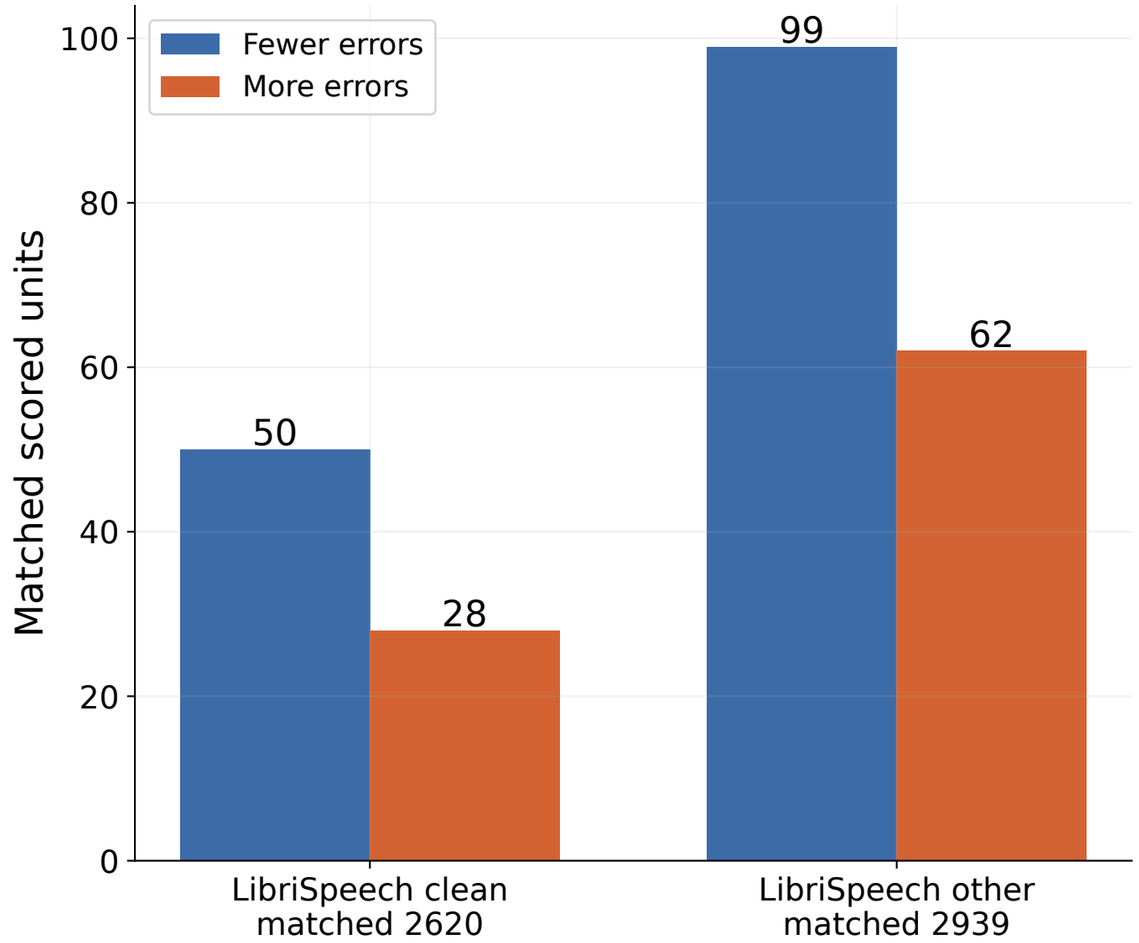




10 observations per run per component; arithmetic means. Core score has a different divisor and is not a like-for-like quality metric.

Matched examples quantify improvements and regressions





Step600 full dumps; unchanged stable IDs and references. Earnings grouped341 chunks into6 parents; OpenML393,LS clean2620,LS other2939.

Stable-ID cases: content and scoring-sensitive forms

LibriSpeech clean | ID 1188-133604-0019 | errors 2 → 0

REF: the large letter contains indeed entirely feeble and ill drawn figures that is merely childish and failing work of an inferior hand it is not characteristic of gothic or any other school

L2 T1: The large letter contains indeed entirely feeble and I'll drawn figures that is merely childish and failing work of an inferior hand. It is not characteristic of Gothic or any other school.

BCR16: The large letter contains indeed entirely feeble and ill drawn figures. That is merely childish and failing work of an inferior hand. It is not characteristic of Gothic or any other school.

Source: matched_dumps.json / ls_clean; baseline line233, target line233; step600

LibriSpeech clean | ID 3729-6852-0046 | errors 0 → 2

REF: … of everything she never allowed him to be in need of anything and she gave no account of his money which she kept altogether because he never asked her to render any accounts

L2 T1: … everything. She never allowed him to be in need of anything, and she gave no account of his money, which she kept altogether because he never asked her to render any accounts.

BCR16: … rything. She never allowed him to be in need of anything, and she gave no account of his money, which she kept all together, because he never asked her to render any accounts.

Source: matched_dumps.json / ls_clean; baseline line724, target line724; step600

Excerpts are illustrative; complete outputs, references and logged error counts retained in evidence.json. No acoustic cause inferred.

Stable-ID cases: content and scoring-sensitive forms

LibriSpeech other | ID 6070-86745-0002 | errors 4 → 0

REF: at a quarter to ten a valet entered he composed with a little groom named john and who only spoke english all albert's e …

L2 T1: At 9:45 a valet entered. He composed with a little groom named John, and who only spoke English, all Albert's establishm …

BCR16: At a quarter to ten a valet entered. He composed with a little groom named John, and who only spoke English, all Albert' …

Source: matched_dumps.json / ls_other; baseline line1591, target line1591; step600

LibriSpeech other | ID 2609-156975-0016 | errors 8 → 12

REF: was moses justified in resisting the egyptian taskmaster

L2 T1: Which mores it's justified and resists the Egyptian tax master.

BCR16: Which mo's it's just findin' resists in the gipsy taks, Master?

Source: matched_dumps.json / ls_other; baseline line398, target line398; step600

Excerpts are illustrative; complete outputs, references and logged error counts retained in evidence.json. No acoustic cause inferred.

Stable-ID cases: content and scoring-sensitive forms

OpenML bad | ID 48239893859244850 | errors 3 → 0

REF: Um jantar simples e popular, especialmente no verão, é o Pa amb Oli: pão com azeite de oliva, tomate e qualquer ingrediente à disposição, como queijo, atum, etc.

L2 T1: Um jantar simples e popular, especialmente no verão, é o "pain avec olive", pão com azeite de oliva, tomate e qualquer ingrediente à disposição, como queijo, atum, etc.

BCR16: Um jantar simples e popular, especialmente no verão, é o "pa amb oli", pão com azeite de oliva, tomate e qualquer ingrediente à disposição, como queijo, atum, etc.

Source: matched_dumps.json / openml_bad; baseline line135, target line135; step600

OpenML bad | ID 11746087412974645092 | errors 2 → 6

REF: Angel 2006 esclarece a abordagem Continuum como uma metodologia usado para ajudar as organizações a alcançarem níveis mais altos em suas performances.

L2 T1: Engel, 2006, esclarece a abordagem continuum como uma metodologia usada para ajudar as organizações a alcançarem níveis mais altos em suas performances.

BCR16: Engel (2006) esclarece a abordagem continuum como uma metodologia usada para ajudar as organizações a alcançarem níveis mais altos em suas performances.

Source: matched_dumps.json / openml_bad; baseline line121, target line121; step600

Excerpts are illustrative; complete outputs, references and logged error counts retained in evidence.json. No acoustic cause inferred.

Earnings: six parents, small net change hides mixed effects

Full matched scope

All341 chunks form6 scored parent recordings,each counted once. Two parents improve,four regress; net+3 word errors. Official p_err4.4560% →4.4755%(+0.0195pp).

Largest parent regression

Stable parent c6efdf26-e19a-4e33-8bbc-c40283fea3a3.mp3:65 →74 errors(+9),step600. Example: baseline “open the call to Q&A” vs target “open a call Q&A”; these are excerpts from one parent,not nine independently diagnosed causes.

Largest parent improvement

Parent03c13d67-c016-4a2b-91d8-e013d5d6bd2b.mp3 improves by11 errors. Mixed parent-level directions make a blanket Earnings regression claim too broad.

Source and limit

Complete parent references,chunk line lists,raw outputs and grouped clean outputs are in evidence.json →examples →earning_chunk_verb →all_pairs. Text inspection alone cannot identify acoustic causes.

Pinned source audit and collected W&B configs; live stop verified by exact Ray submission.

Check scoring-sensitive changes before tuning rewards

Evidence

At step600: LS clean−24 errors,LS other−42,Earnings+3,OpenML−2 net errors. OpenML has2 samples with changed n_ref,so its corpus ratio needs both denominators. Table snapshots match20/20 examples per run; the full dumps cover datasets absent from selected tables.

Bounded next check

Pinned scorer replay reproduces all12 selected cases. In OpenML ID1174608741297464509 2,parenthesized(2006) disappears from the normalized hypothesis: errors2 →6. Audit bracket/number normalization first; retain official scores and test a symmetric diagnostic convention on both runs.

Success criteria

Keep official per-dataset p_err unchanged in the report. Any later one-factor trial should fix divisor/exposure confounds,reduce verified scoring-pattern errors at matched budget,and guard LS clean,LS other,OpenML and Earnings separately. No new training was launched.

Pinned source audit and collected W&B configs; live stop verified by exact Ray submission.